

ESP Pumps

Operations, AI Modelling & Assistant Development

Introduction deck for software-background AI interns joining the ADVAIT ESP Asset Performance Monitoring & Management program

Equipment domain → data model → AI advisor

Build useful AI on trustworthy plant data

Intern outcome

Source visuals/data: uploaded ESP knowledge base; SLB Defining Series.

Electrical Submersible Pumps

Rick von Flatern
Senior Editor

More than 90% of all producing oil wells require some form of artificial lift to increase the flow of fluids from wells when a reservoir no longer has sufficient energy to naturally produce at economic rates or to boost early production to improve financial performance. One of the most versatile and adaptable artificial lift methods is electric submersible pumping.

Deployed in an estimated 150,000 to 200,000 wells worldwide, electric submersible pumps (ESPs) consist of multiple centrifugal pump stages mounted in series within a housing mated to a submersible electric motor. These pumps are connected to surface controls and electric power by armor-protected cables. Surface electric drives power and control ESP systems, which are able to lift from 16 to 4,770 m³/d [100 to 30,000 bbl/d], a pump rate operation range that surpasses the performance of other pump-type artificial lift systems such as rod pumps, progressing cavity pumps and hydraulic lift.

ESP Origins

In 1911, 18-year-old Russian engineer Armais Arutunoff invented the first electric motor that operated in water. He added a drill and a centrifugal pump to the motor, inventing what is now known as the electric submersible pump. Arutunoff immigrated to the US, where he founded Russian Electrical Dynamo of Arutunoff, or REDA. Electric submersible pumps are now operating in onshore and offshore fields throughout the world.

Anatomy of an ESP System

Electric submersible pump systems comprise both downhole and surface components (Figure 1). The overall length and diameter of ESP downhole equipment are designed based on the horsepower necessary to deliver the desired flow rate.

An ESP is a multistage centrifugal pump whose stages are stacked; the operating requirements of the well and completion design dictate the number of stages. Each stage contains a rotating impeller and stationary diffusers typically cast from high-nickel iron to minimize abrasion or corrosion damage. As well fluid flows into the first stage of the ESP, it passes through an impeller, and the fluid is centrifuged radially outward, gaining energy in the form of velocity. The centrifugal pump is driven by an induction motor that can attain operating speeds of more than 5,000 rpm when using a variable speed drive.

After it exits the impeller, the fluid is forced to make a sharp turn to enter the diffuser. As it travels through this section, the fluid is diffused, and its velocity is converted to pressure. The fluid, which now has a slightly higher pressure than when it entered the first stage, enters the next impeller and diffuser stage to repeat the process. The fluid passes through all stages of the pump, incrementally gaining pressure in each stage until it achieves a total developed head, or designated discharge pressure, and has sufficient energy to travel to the surface of the well.

A pump's operational performance is illustrated in its pump performance curve (Figure 2). A performance curve is generated for each pump

Oilfield Review 2015.

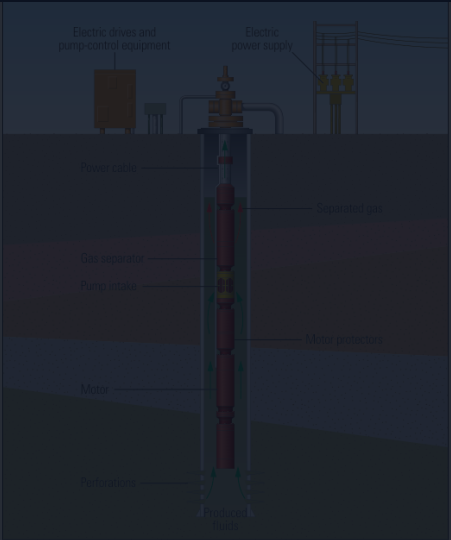


Figure 1. Typical ESP configuration. Downhole ESP components include motor, protectors, pump sections, pump intakes, power cables, gas-handling equipment and downhole sensors (not shown). Surface components include pump-control equipment such as variable-speed drives and an electric power supply.

manufactured and quantifies the relationship between pump horsepower, efficiency, flow rate and head relative to the operating flow rate. The pump recommended operating range is defined for each pump stage in the catalog

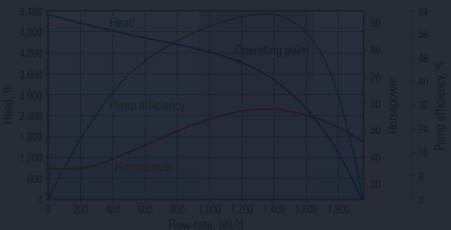
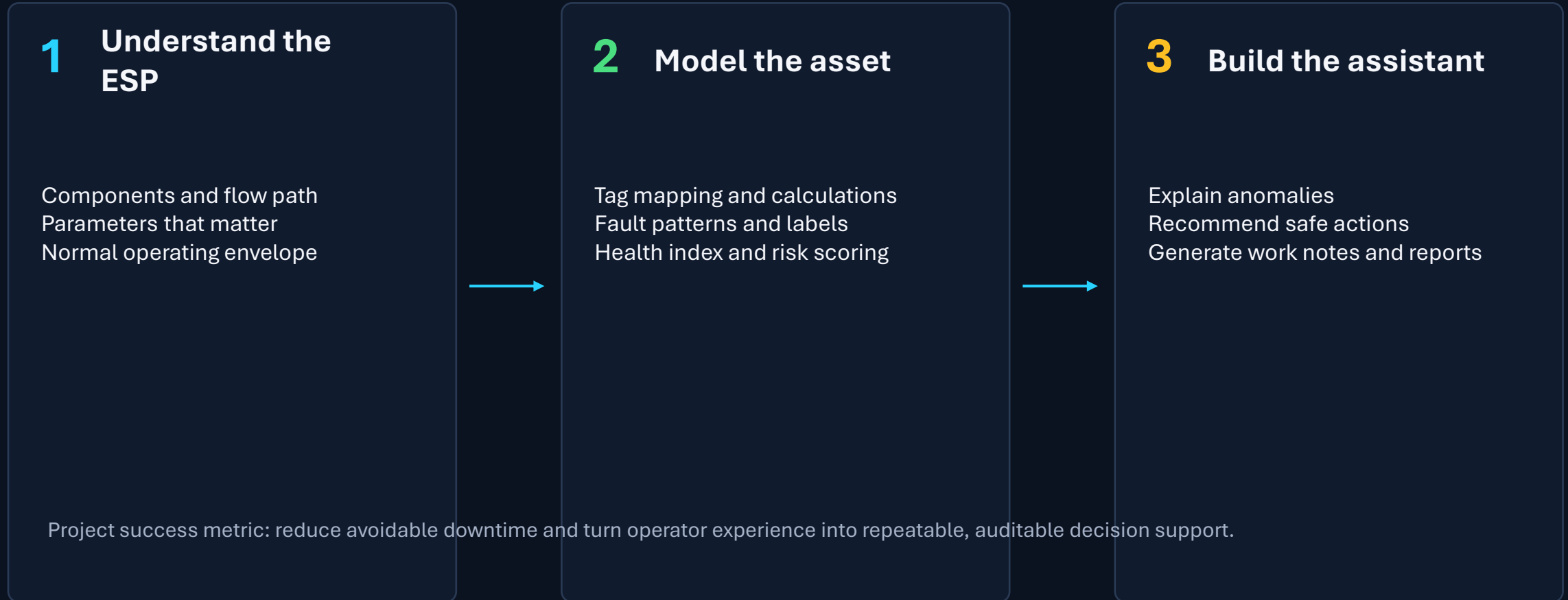


Figure 2. Pump performance curve. Such curves are generated to chart a pump's ability to displace fluids and to determine the number of pump stages required to achieve a desired production rate. The head capacity curve (blue) shows the amount of lift at a given flow rate. The horsepower requirements of the pump (red) across a range of flow capacities are derived from performance testing. The pump efficiency (green) is calculated from the head, flow capacity, fluid specific gravity and horsepower. The purple shaded area indicates the most efficient operation range for this

What interns need to deliver

Software capability only becomes valuable when it respects equipment physics and operating reality.



Why ESP exists in oil production

When reservoir energy is not enough, artificial lift supplies the missing pressure to move fluids to surface.

ESP = electric motor + multistage centrifugal pump downhole

High-volume artificial lift for wells that cannot flow economically by natural reservoir pressure.

Typically selected for moderate-to-high liquid rate wells where rod-pump or gas-lift economics are weak.

The system converts electrical energy into hydraulic head inside the well.

Software analogy: ESP is a cyber-physical production service with hard constraints

Life Factors

Run life of an ESP is based on many factors

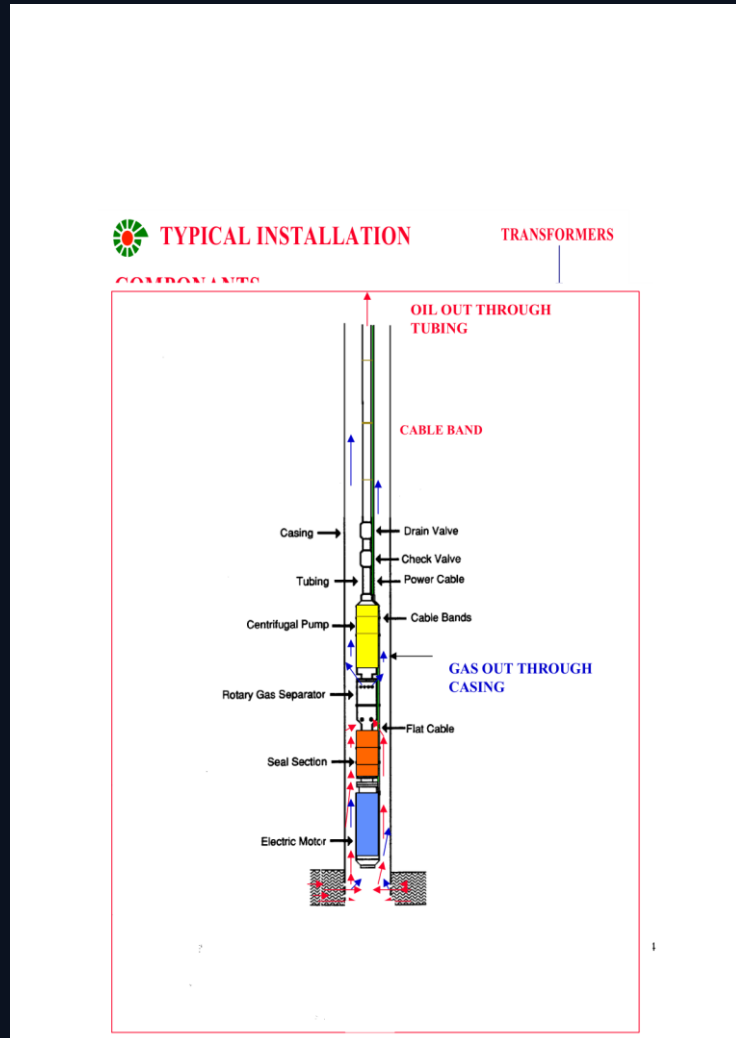
Wells are different and may have a combination of factors that affect ESP run life

Run life will be determined by the limiting factor in the well

Centrilift

ESP system anatomy

Separate the asset into surface power/control, downhole hydraulic string and well/reservoir context.



Downhole assembly

Motor
Protector / seal section
Intake or gas separator
Multistage pump
Cable, check valve, drain valve
Downhole gauges

Surface system

Transformer
VSD / switchboard
Junction box
Wellhead
SCADA / RTU / historian
Power quality monitoring

How the ESP actually pumps

Each stage adds a small pressure increase; many stages stacked together create the total developed head.

Stage-level physics

1

Fluid enters the impeller from pump intake.

2

Impeller accelerates fluid outward, adding velocity.

3

Diffuser converts velocity into pressure.

4

Stages repeat until discharge pressure is achieved.

This motion is caused by centrifugal force. The other motion moves in direction tangential to the outside diameter of the impeller. As a result of these two components is the actual direction of flow. The diffuser's function is to change the velocity energy into pressure energy.

There are two general types of design for submersible pumps. The smaller flow pumps are mostly of radial flow design. Figure 2-a shows such a stage configuration. In this type the impeller discharges the fluid in mostly a radial direction. As the pumps reach design flows of approximately 300 M3/d in the 400 series pumps and 550 M3 /d in the larger diameter pumps, the design changes to a mixed flow. Figure 2-b shows this configuration. The impeller in this stage design imparts a direction to the fluid that contains substantial axial direction as well as radial direction.



ELECTRICAL SUBMERSIBLE PUMP (ESP)

CENTRIFUGAL PUMP STAGES

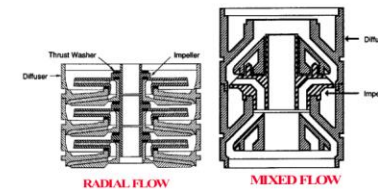


Figure 2-a Figure 2-b

In many of the pump designs, the impeller is free to float axially on the shaft and the individual impeller stage is absorbed on specially designed pads found on the diffuser. A specially designed thrust bearing contained in the seal section carries only the thrust of the pump shaft. This configuration is called a floating stage design. The benefit of this design is that many stages can be stacked together without having to fix the impellers axially on the shaft with precise alignment. As a result, pumps can be manufactured having hundreds of individual stages.

7

Pump curve is the operating contract

The AI model must know whether the current point is inside the recommended operating range, not just whether values are high or low.

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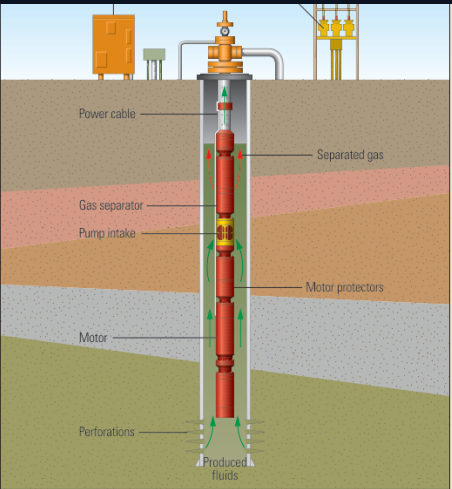


Figure 1. Typical ESP configuration. Downhole ESP components include motor, protectors, pump sections, pump intakes, power cables, gas-handling equipment and downhole sensors (not shown). Surface components include pump-control equipment such as variable-speed drives and an electric power supply.

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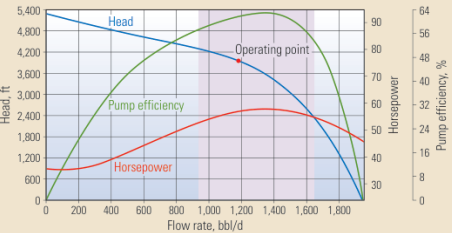


Figure 2. Pump performance curve. Such curves are generated to chart a pump's ability to displace fluids and to determine the number of pump stages required to achieve a desired production rate. The head capacity curve (blue) shows the amount of lift at a given flow rate. The horsepower

What the curve tells us

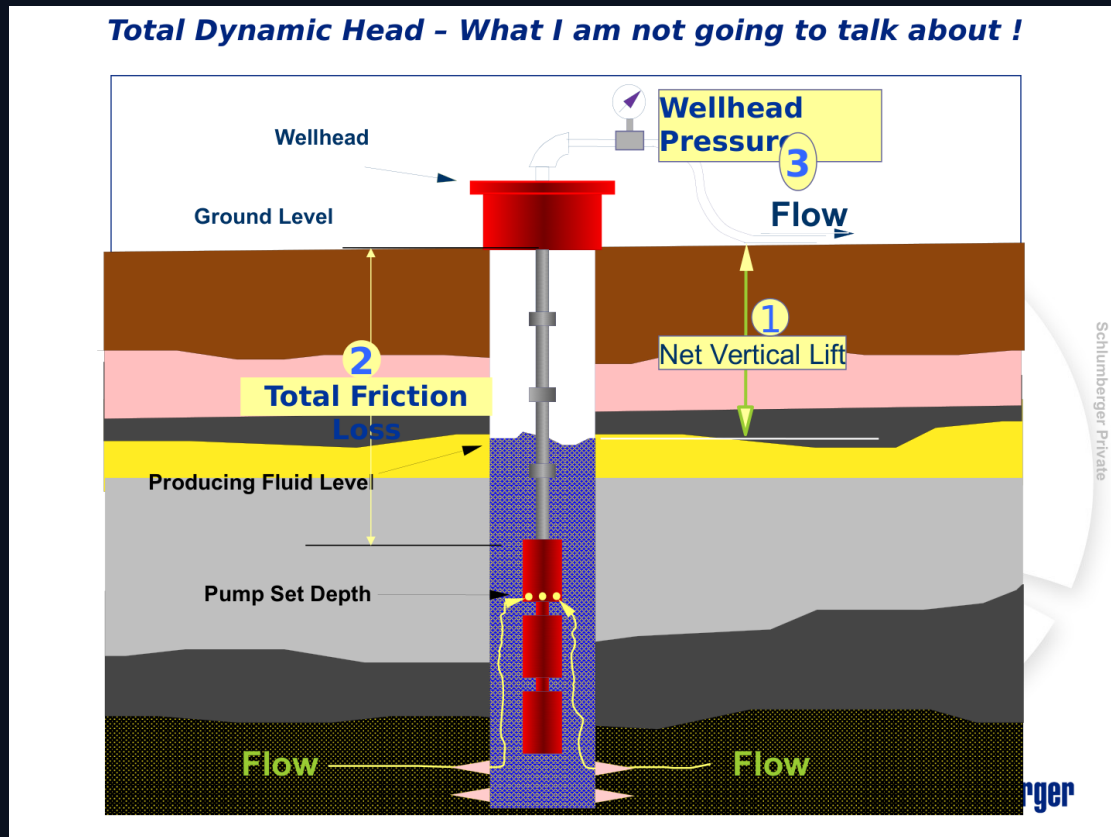
- Flow rate vs head: hydraulic capability
- Horsepower: motor load requirement
- Efficiency: best operating zone
- Operating point: actual production reality
- Left of range: downthrust/low-flow risk; right of range: upthrust/high-flow risk

Core AI feature: deviation from design / BEP / ROR

Source: SLB Defining Series page 1; pressure-gradient design slides.

Total Dynamic Head links well conditions to pump demand

TDH is where production engineering meets sensor data.



TDH = vertical lift + friction + wellhead pressure

Net vertical lift changes with producing fluid level.
Tubing friction rises with flow rate and fluid properties.

Wellhead pressure/backpressure shifts the required discharge pressure.

Wrong density, water cut or GOR assumptions break the model.

Implementation note: create a TDH calculation service, keep every assumption visible, and compare calculated PDP/PIP with gauge values.

Minimum tag universe for useful monitoring

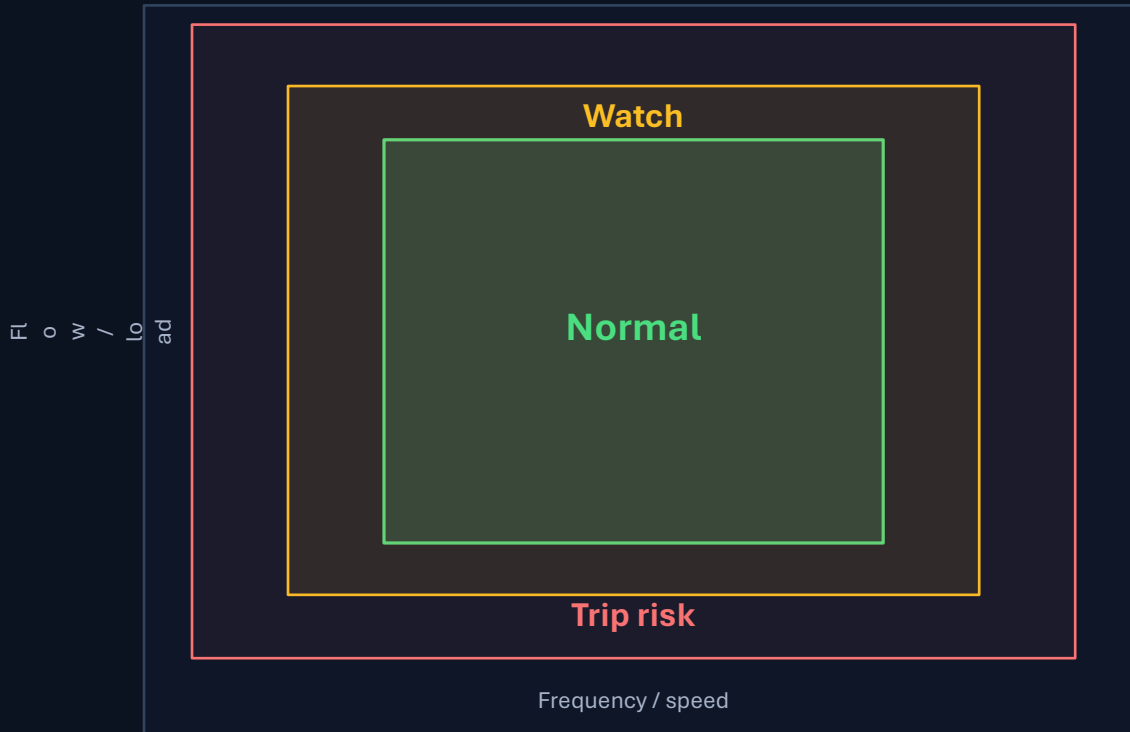
Interns must convert raw historian tags into physics-aware features and asset-level state.

Domain	Typical tags	Why it matters
Production	Flow rate, WHP, choke, tubing/casing pressure	Confirms production and hydraulic loading
Downhole	PIP, PDP, motor temp, intake temp, vibration	Early signatures of gas, wear, blockage, overheating
Electrical	Voltage, current, amps imbalance, frequency, leakage	Motor health and power quality
Design static	Pump model, stages, ROR, motor HP, cable, set depth	Benchmark for deviation and safe envelope
Context	Water cut, GOR, API, BHT, PI, reservoir pressure	Explains why same symptoms differ by well

Source: pressure-gradient measurements list; real-time monitoring docs.

Operating envelope: monitor, explain, act

The application should distinguish “needs watch” from “stop now” with traceable logic.



Normal: keep speed, maintain surveillance.
Watch: explain deviation and recommend controlled parameter adjustment.
Warning: create exception, request engineering review.
Critical: trip prevention or safe shutdown workflow.

Never allow an AI assistant to command unsafe action directly

Run life is limited by the weakest operating factor

Failure is rarely “one variable high”; it is usually a context-dependent combination.

Common Run Life Factors

Wrong Sizing of Equipment
High (BHT) Temperature
Free Gas
Viscosity
Corrosion
Sand / Foreign Material Production
Position Tendencies
Electrical Failures
Operational Problems
Equipment Age



Centrilift

Key degradation drivers

Wrong sizing or operating outside ROR
High bottom-hole / motor temperature
Free gas and gas lock / gas interference
Sand / solids abrasion and erosion
Corrosion and scale deposition
Electrical failures and power quality problems
Operational cycling, frequent starts/stops

Fault taxonomy for modelling and assistant logic

Start with failure modes operators recognize, then map them to symptoms and data signatures.

Hydraulic

Gas interference
Intake blockage
Stage wear
Tubing leak

Mechanical

Broken shaft
Bearing wear
Vibration
Misalignment

Electrical

Overload
Underload
Phase imbalance
Insulation leakage

Thermal

Motor overheating
Poor cooling
High BHT
High starts/stops

Well / process

Reservoir pressure
change
Water cut shift
Sand production
Scale/corrosion

Use this taxonomy in the database, label schema, model output class names, exception workflows and assistant retrieval corpus.

Troubleshooting is pattern recognition with physics

The same symptom can mean different faults: use multiple parameters together

ChampionX

ARTIFICIAL LIFT



ESP Troubleshooting Guide

	Parameter					
	Flow Rate	Wellhead Pressure	Operating Amps	Pump Discharge Pressure	Pump Intake Pressure	Motor Temperature
Broken Shaft	Reduced or No Flow	Low or Facility Pressure Only	Lower or Idle Motor Amps	Lower or equal to Intake Pressure	Increasing	Increasing High Temperature
Hole in Tubing	Reduced or No Flow	Low or Facility Pressure Only	Steady or Increasing	Decreasing	Increasing	Increasing High Temperature
Blockage at Pump Intake	Reduced or No Flow	Low or Facility Pressure Only	Lower or Idle Motor Amps	Lower or equal to Intake Pressure	Increasing	Increasing High Temperature
Blockage at Perforations	Reduced Flow	Decreasing	Decreasing	Steady or Increasing	Steady or Decreasing	Increasing
Start-up with Kill Fluid	Steady or Increasing	Reduced	Elevated	Elevated	Steady	Elevated
Shut in at Surface	No Flow	Excessive	Lower or Idle Motor Amps	Elevated	Increasing	Increasing High Temperature
Blockage in Pump Stage	Reduced or No Flow	Reduced	Reduced	Reduced	Increasing	Increasing High Temperature
Increase in Reservoir Pressure	Steady or Increasing	Elevated	Elevated	Steady or Reduced	Increasing	Steady or Reduced
Presence of Free Gas at Pump Intake	Reduced or No Flow	Reduced	Erratic	Erratic	Increasing	Increasing High Temperature
Stage Wear	Reduced Flow	Decreasing	Elevated	Reduced	Elevated	Increasing
Increase in Frequency	Increasing	Elevated	Elevated	Elevated	Reduced	Elevated

UNBRIDLED ESP Systems values your business and the opportunity we earn to serve you. Don't hesitate to contact a member of the team at any time. We are committed to providing exceptional service throughout the life of your asset.

	Contact	Phone	Email
Sales	Permian Sales Main Number	918-396-0558	
Service	Permian Field Service	432-258-5362	Blake.Bartosh@championx.com

Example pattern families

Reduced/no flow + rising PIP + high temperature: blockage, broken shaft or shutdown-at-surface scenarios.

Erratic amps and pressure with reduced flow: free gas at pump intake.

Reduced flow + elevated amps + reduced discharge pressure: possible stage wear.

Increasing frequency should raise flow, WHP, amps, PDP and temperature unless constrained.

Source: ChampionX/Unbridled ESP Troubleshooting Guide page 1.

High-value fault signals: vibration, current, temperature

For machine learning, high-frequency mechanical/electrical features can detect faults earlier than slow process tags.



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Electric submersible pump vibration analysis under several operational conditions for vibration fault differential diagnosis

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ARTICLE INFO

Keywords:
Vibratory behavior
Synchronous frequency variation
Differential diagnosis
Correlation analysis
Temperature influence

ABSTRACT

An Electric Submersible Pump (ESP) vibration analysis was performed, considering different wear states and operational conditions. The pumps were tested with different fluid viscosities, operating points, and speeds to evaluate their vibration behavior, with the aim of providing characteristics for non-invasive differential vibration diagnosis. A specific frequency spectrum estimation method is described, focusing on the extraction of frequency component vibration amplitudes used in the petroleum industry-standard vibration analysis. A time-interval definition procedure was proposed to reduce signal amplitude losses due to variation of synchronous operating frequency. The results indicate that the frequency component peak amplitudes can be more accurately identified during synchronous frequency variation by the proposed method than by typical estimation methods. In this study, an ESP vibration differential diagnosis was achieved by analyzing relations between orders of the synchronous frequency, and by distinguishing that the synchronous component amplitude rose approximately proportionally to the square of the rotating speed; this differentiates unbalance fault from a bent shaft or a misalignment. A correlation matrix analysis is provided to demonstrate that variation in the fluid-temperature difference between the pump intake and discharge is related to the vibration amplitude variation in a pump with a vibration fault.

1. Introduction

Electrical submersible pumps (ESPs) are the second most common artificial lifting method applied worldwide (Liang et al., 2015), deployed in, it is estimated, 150,000 to 200,000 wells (Flatern, 2015). Electrical submersible pumps account for approximately 10% of the world's crude oil production (Takacs, 2017). These systems are ideally suited to pumping high volumes at high pressure, and the petroleum industry applies the method to pump petroleum at a high flow rate in offshore applications.

An electrical submersible pump is composed of several small-diameter centrifugal stages (typically 4–9 in.) that are mounted serially, with an operational frequency range varying from 30 Hz to 60 Hz. In a submerged ESP installation, a multistage centrifugal pump is coupled to a magnetic induction motor using a protector seal assembly filled with an insulating fluid that is heavier than water. The electric motor is cooled by the oil-well fluids that pass through the motor housing. For subsea systems, the output motor power may be higher than 1000 hp. The structure can reach more than 40 m in length, and the pump can operate in a high-temperature environment, pumping fluids containing abrasives (Minette et al., 2016).

The installation and intervention costs of ESP-based pumping systems are usually higher than those of other elevation methods, particularly in the case of deep-sea, underwater wells (Ribeiro et al., 2005). In addition to the high costs of installation and intervention, faults in this equipment usually cause significant production losses since they typically operate in high-production petroleum wells. A careful evaluation of ESP systems before installation is critical to prevent premature operational failure.

The risk of vibration problems is increased in equipment such as ESP systems due to the difference between its considerable length and its small axial diameter. The factory acceptance evaluation process of the ESP system for fault diagnosis includes an expert vibration analysis of collected accelerometer signals during a test well operation, following

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Feature direction

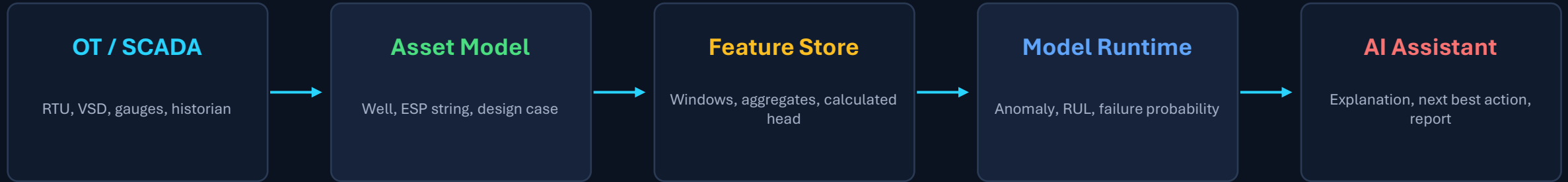
Order tracking around shaft speed: 1X, 2X, harmonics and sidebands.
RMS, peak, crest factor, kurtosis and band energy from vibration windows.
Current signature features: imbalance, THD proxy, underload/overload, leakage trend.
Temperature rise vs load and cooling flow as an early thermal risk indicator.

ML caution: do not train only on “pretty” data; include startups, trips and sensor dropouts

Source: ESP vibration analysis paper and uploaded troubleshooting guides.

ADVAIT ESP-PMM data pipeline

The backend must preserve raw evidence while publishing clean features for models and assistant reasoning.



Critical principle: lineage from recommendation back to raw tags, design data and calculation version is non-negotiable.

Feature engineering backlog

A strong ESP model starts with robust features before chasing complex algorithms.

Hydraulic calculated

$\Delta P_{\text{pump}} = \text{PDP} - \text{PIP}$
Head = $\Delta P / \text{gradient}$
TDH vs pump curve
PIP drawdown vs IPR

Operating behavior

Frequency-normalized flow
Start/stop count
Time outside ROR
Trend slopes and rate of change

Thermal/electrical

Motor temp margin
Current vs frequency
Underload/overload events
Voltage/current imbalance

Data quality

Sensor stale flag
Dropout interpolation
Unit conversion confidence
Tag binding health

Source: pressure-gradient calculation slides and monitoring parameter lists.

AI modelling workstreams

Build models in layers: rules for transparency, ML for prediction, retrieval for assistant context.

Layer	Purpose	Candidate methods	Intern tasks
Rule engine	Known physics and operating limits	Thresholds, state machines, pump-curve checks	Encode ROR, TDH and trip logic
Anomaly detection	Detect abnormal behavior without labels	Isolation Forest, autoencoder, robust z-score	Build baseline by well/pump type
Failure prediction	Estimate probability and lead time	Gradient boosting, temporal CNN/LSTM, survival model	Create labels and evaluate lead time
Fault classification	Identify likely root cause	XGBoost, random forest, sequence classifier	Map symptoms to fault taxonomy
RAG assistant	Explain and recommend	Retriever + tool calling + evidence cards	Build prompt, citations, action templates

Source: Siemens OTC AI predictive maintenance paper and uploaded troubleshooting papers.

Prediction without labels is theatre

Define the exact event, prediction horizon and business metric before model training.

Event labels

- Trip event
- Failure pull/workover
- Operating exception
- Avoided failure / intervention
- Root cause after teardown

Time horizons

- Next 1 hour: trip risk
- Next 24 hours: severe exception
- Next 7–14 days: failure risk
- Remaining Useful Life estimate

Metrics

- Lead time before failure
- False alarm rate per well-month
- Precision on critical alarms
- Production deferment avoided
- Operator action adoption

The Siemens ESP pilot reported failure probability forecasting 12 days before an actual failure; use that as an ambition, not as a guaranteed benchmark.

ESP AI Assistant: what it must do

The assistant is not a chatbot; it is an evidence-backed operational copilot.

Observe

Read asset state, data quality, recent trends and active exceptions.

Diagnose

Match multi-signal patterns to likely failure modes and confidence.

Explain

Show evidence: tag movements, calculations, design envelope and source procedure.

Recommend

Give safe next steps: monitor, adjust, inspect, escalate or plan intervention.

Document

Generate shift notes, exception reports and model-feedback labels.

Every answer should include confidence, evidence and escalation criteria

Guardrails for industrial AI

Wrong advice can damage equipment or stop production. Build controls from day one.

Assistant must never

- Invent missing sensor values
- Recommend unsafe speed changes without constraints
- Suppress alarms because model confidence is low
- Hide uncertainty or source evidence
- Override operator/company procedures

Assistant must always

- State confidence and data quality
- Link recommendation to tags/calculations
- Offer escalation route
- Create audit trail
- Support human approval before action

Industrial AI = bounded autonomy + human accountability

Implementation backlog for interns

Each task should create reusable product capability, not isolated notebooks.

Work item	Output	Definition of done
Asset schema	ESP asset definition tables + sample data	Can configure well, pump, motor, protector, cable, VSD and design case
Tag binding	Historian tag dictionary + unit normalization	Raw tags map to canonical ESP variables with data-quality flags
Calculation service	TDH, ΔP , head, ROR deviation, temp margins	Values reproducible and versioned with assumptions
Fault rules	Rule cards for 10–15 common patterns	Each rule has trigger, evidence, severity and recommended action
ML baseline	Anomaly and failure-risk prototype	Backtest notebook + API-ready scoring function
Assistant prototype	RAG + tool calling over ESP knowledge base	Answers cite evidence and produce action-ready summary

Suggested 6-week onboarding execution plan

Ship a vertical slice that can be demonstrated with synthetic + historian-ready data.



Demo target: select one ESP well → show current operating point → identify anomaly → explain likely cause → generate recommended action note.

Intern deliverables and quality bar

Good interns will produce product-grade assets, not just experiments.

Data

Canonical variables
Sample datasets
Data quality checks

Engineering

Calculation APIs
Rule engine
Test cases

AI

Baseline models
Evaluation reports
Prompt/RAG harness

Product

Screens/demo flow
Exception workflow
User documentation

Quality bar: reproducible, versioned, explainable, testable and demo-ready

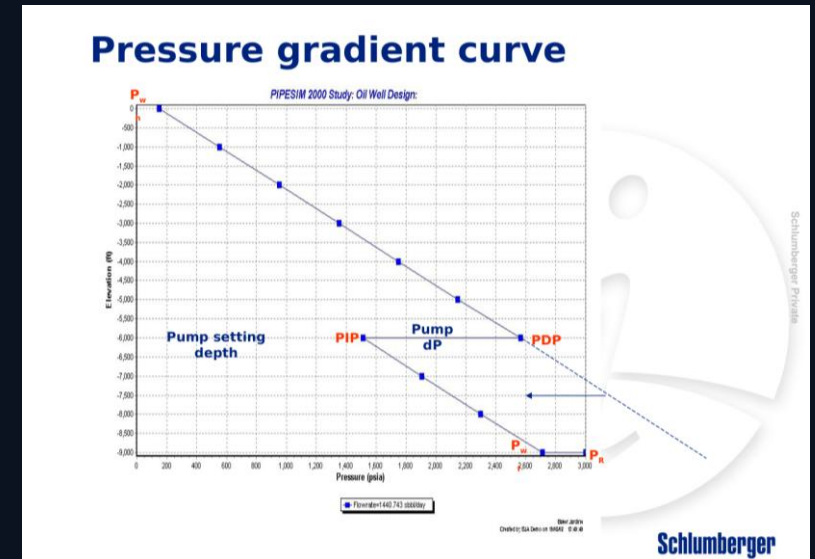
Recommended reading map

Use the sources below as the knowledge backbone for intern onboarding and RAG ingestion.

Topic	Primary internal/public source
ESP basics and configuration	SLB Defining Series: Electrical Submersible Pumps; ESP Manual introduction
Components	ESP Components Equipment Description; ESP system training decks
Design and TDH	Pressure-gradient curves and ESP Total Dynamic Head design notes
Operations and run life	ESP Run Life Factors; ESP Basic Design and Operational Factors
Troubleshooting	SPE-199091-MS troubleshooting guide; combined troubleshooting matrix
AI and predictive maintenance	OTC-29384-MS Siemens AI use case; vibration analysis paper
Assistant build requirements	ADVAIT ESP-PMM project context and current module requirements

Intern mission

Turn ESP data into actionable intelligence — with physics, evidence and operational discipline built in.



Best first demo: one well, one current operating point, one likely fault, one traceable recommendation.